Simple Calculator using C

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1. Operating Manual

1.1 Two Modes (Interactive Mode and CLI Mode)

The calculator supports two execution modes:

- **Interactive Mode**: Simply run `./calculator`. You will enter a REPL (Read-Eval-Print Loop) where you can type expressions one by one. Type `quit` to exit.
- **CLI Mode**: Run `./calculator "1 + 2 * 3"`. The calculator will evaluate the expression provided as an argument and exit immediately.

1.2 Parameter Setting & Formatting

Use the following flags to customize the calculation and output behavior:

1.2.1 Calculation Precision (`-p`)

Use the `-p` flag to specify the number of decimal places for the core calculation and default output (default is 6).

- **Example**: `./calculator -p 10 "sqrt(2)"`
- **In Interactive Mode**: You can also type `-p 10` during the session to update the precision for all subsequent calculations.
- **Syntactic Sugar**: You can use `-p max` to safely push the precision to the engine's absolute safe limit (e.g., ~3000 decimal places), avoiding manual buffer limit calculations.

1.2.2 Output Format & Scientific Notation (`-x`, `-s`)

The calculator dynamically supports formatting large or infinitesimal numbers into scientific notation (e.g. `1.23e+10`).

- **Notation Mode (`-x`)**: Toggles the display mode.
  - `-x auto` (Default): Automatically switches between fixed-point and scientific notation based on the number's magnitude (similar to C's `%g`).
  - `-x off`: Forces fixed-point notation for all numbers, completely unrolling decimals.
  - `-x on`: Forces scientific notation for all numbers.
- **Scientific Precision (`-s`)**: Sets the number of decimal places to keep in the mantissa when outputting in scientific notation (default is 6).
  - **Example**: `./calculator -p max -x on -s 3 "7 / 0.3"` → `2.333e+01`

1.3 Input Format

The calculator supports:

- **Numbers**: Including very large integers, decimals and Scientific notation (e.g., `1234567890.12345`, `1.23e4`, `1.23E4`).
- **Operators**: `+`, `-`, `*`, `/`.
- **Grouping**: Parentheses `(...)` to override operator precedence.
- **Functions**: `sqrt(x)` for square root and `abs(x)` for absolute value.

2. Basic Functions

2.1 Large Number Calculation (BigInt)

Standard C integer types (`int`, `long`) overflow for numbers beyond 64 bits. To support arbitrarily large numbers, all values are stored and computed as **null-terminated character strings**, with arithmetic implemented digit-by-digit — essentially the manual column arithmetic we learned in school.

Core principle:

1. **Represent numbers as strings** — `"1234567890123456789012345678 90"` never overflows.
2. **Process digits from the least significant end** — reverse both strings, iterate position by position, sum/multiply digits and propagate carry.
3. **Handle signs separately** — strip the leading `-`, determine the result sign by the combination of operands' signs, then perform unsigned (`abs`) arithmetic on the digits.

Code structure for addition (`bigint_add`):

```
void bigint_add(const char *num1, const char *num2, char *result) {
    // 1. Detect signs, strip '-' prefix
    // 2. Same sign  → bigint_abs_add(), prepend sign if negative
    // 3. Mixed sign → compare magnitudes → bigint_abs_sub(larger,
smaller)
    //                 prepend '-' if the negative operand was larger
}
```

```
static void bigint_abs_add(const char *num1, const char *num2, char
*result) {
    // Iterate from length-1 down to 0, summing digits + carry
    // Write result in reverse, then reverse_string()
}
```

The same pattern — sign detection → unsigned core → sign restoration — applies to `bigint_sub`, `bigint_mul`, and `bigint_div`.

2.2 Decimal Calculation

The BigInt library from section 2.1 operates on **pure integers only** — it has no concept of a decimal point. To support decimal numbers, three lightweight helper functions were introduced as a "decimal layer" that wraps around the BigInt core:

Function	Role
`extract_decimal(input, out, scale)`	Strips the `.` from `"3.14"` → `"314"`, records `scale = 2`
`append_zeros(str, n)`	Pads `n` trailing zeros to align two integers to the same scale before add/sub
`insert_decimal(str, scale)`	Inserts `.` back at position `scale` from the right: `"464"` → `"4.64"`

How they work together in `apply_operator`:

```
// Example: 3.14 + 1.5
extract_decimal("3.14", int1, &scale1);  // int1="314", scale1=2
extract_decimal("1.5",  int2, &scale2);  // int2="15",  scale2=1

int max_scale = max(scale1, scale2);     // max_scale=2
append_zeros(int2, max_scale - scale2);  // int2 → "150"  (align to same
scale)

bigint_add(int1, int2, result);          // "314" + "150" = "464"
insert_decimal(result, max_scale);       // "464" → "4.64"
```

For **multiplication**, no alignment is needed — the result's scale is simply `scale1 + scale2`. For **division**, the alignment is the same as add/sub, but `bigint_div` handles the decimal output internally via the `precision` parameter.

2.3 Manual Precision (`-p`)

2.3.1 Dynamic Precision Configuration

The system uses a default precision of 6 decimal places, aligning with the standard `bc` utility. Precision is controlled by a single `int precision` variable that flows as a parameter through every layer of the

system:

```
main()                              // parses -p N, stores as local
precision
  └─ run_interactive_mode(precision)  // OR run_cli_mode(..., precision)
       └─ evaluate_expression(expr, precision)
            └─ apply_operator(..., precision)
                 └─ bigint_div(..., precision, ...)
                 └─ format_fixed_precision(result, ..., precision)
```

In **interactive mode**, the precision can also be changed mid-session with the `-p` command, which updates the local variable for all subsequent calculations in that session.

2.3.2 Rounding (Half-Up)

Rounding is applied at two points:

In `bigint_div` — after computing `precision` decimal digits, it computes one extra digit to use as the rounding digit:

```
// Compute one extra digit beyond precision
round_digit = div_step(remainder, divisor);
if (round_digit >= 5) {
    // carry +1 into the last kept digit
}
```

In `format_fixed_precision` — trims or pads any intermediate result to exactly `precision` decimal places, with the same half-up rule:

```
int round_digit = frac[precision] - '0';  // look one past the cutoff
frac[precision] = '\0';                   // truncate
if (round_digit >= 5) carry_into(frac);   // round up if needed
```

Edge cases handled: rounding `9.999` at 2 places correctly produces `10.00` by propagating carry into the integer part.

2.3.3 Intermediate Precision Truncation (Accumulation Error)

Because the Shunting-Yard evaluator evaluates complex mathematical equations natively by pushing intermediate strings back onto the Value Stack, every intermediate arithmetic operation undergoes formatting and strict truncation to the specified `-p` calculation precision immediately. As a result, calculations like `1 / (1 / 3)` at `-p 6` will evaluate the inner parenthesis to `0.333333` first, and then `1 / 0.333333` yields `3.000003`. This faithfully mimics the behavior of physical handheld calculators, ensuring that visible intermediate states perfectly and predictably match the final mathematical pipeline without silently carrying hidden fractions.

2.4 Long Expression Evaluation

2.4.1 Shunting-Yard Algorithm

Evaluating complex expressions requires respecting operator precedence and mathematical scopes. We use Dijkstra's **Shunting-Yard algorithm**, using two stacks to process expressions left-to-right:

Stack	Contents
Value Stack	Numbers (BigInt strings) or computed intermediate results
Operator Stack	Operators (`+`, `*`), parentheses `(`, and function markers (`S` for `sqrt`, `A` for `abs`)

The core rule: before pushing a new operator, **pop and apply** all operators on the stack that have **equal or higher precedence**. Parentheses override this logic to enforce scope.

Precedence handling:

Element	Priority / Behavior
Functions (`sqrt`, `abs`)	Pushed to stack as a special single-character marker (`S`, `A`). Evaluated immediately when their enclosing parenthesis closes.
`*`, `/`	Level 2 (High)
`+`, `-`	Level 1 (Low)
`(`, `)`	Parentheses act as boundaries. When `)` is encountered, all operators are popped and applied until the matching `(` is found.

2.4.2 Function Collaboration

The pipeline processing `"sqrt(9) + abs(-5)"` naturally handles functions and brackets:

```
[ Lexer ]                          [ Parser Loop ]                      [
Execution ]
get_next_token()                   evaluate_expression()
apply_operator() / Math APIs

"sqrt"      →  TOKEN_FUNC     →    push 'S' to OpStack
"("         →  TOKEN_LPAREN   →    push '(' to OpStack
"9"         →  TOKEN_NUMBER   →    push "9" to ValStack
")"         →  TOKEN_RPAREN   →    pop operators until '('
                                   If 'S' is underneath, trigger   →
bigint_sqrt("9")
"+"         →  TOKEN_PLUS     →    push '+' to OpStack
"abs"       →  TOKEN_FUNC     →    push 'A' to OpStack
"("         →  TOKEN_LPAREN   →    push '(' to OpStack
"-5"        →  TOKEN_NUMBER   →    push "-5" to ValStack
")"         →  TOKEN_RPAREN   →    pop operators until '('
                                   If 'A' is underneath, trigger   →
abs("-5")
```

```
END          →  TOKEN_END      →   pop remaining '+'                →
apply_operator(3, 5, '+')
```

This dual-stack architecture naturally supports arbitrary nesting like `sqrt(abs(-16) + 9)`.

2.5 Built-in Functions

2.5.1 `sqrt(x)` — Newton's Method on Pure Integers

Rather than introducing floating-point arithmetic, `sqrt` is reduced to a **pure integer problem** using a scaling trick:

$$\sqrt{N} = \frac{\sqrt{N \times 10^{2k}}}{10^k}$$

By appending `2k` zeros to the input's integer representation (via `extract_decimal` + `append_zeros`), we shift the real-valued square root into the integer domain, compute it there, then restore the decimal point with `insert_decimal`.

Newton's iterative formula applied on BigInt strings:

$$x_{n+1} = \frac{x_n + N / x_n}{2}$$

Each iteration calls:

- `bigint_div(N, x, ...)` — to compute `N / x`
- `bigint_add(x, quotient, ...)` — to sum `x + N/x`
- `bigint_div(sum, "2", ...)` — to halve

Termination: by the AM-GM inequality, once the guess overshoots the true root, all subsequent values **monotonically decrease**. Iteration stops when the new value is no longer smaller than the previous one.

Decimal input handling: if the input is a decimal (e.g. `sqrt(2.25)`), `extract_decimal` first strips the decimal point to produce an integer string `"225"` with `scale = 2`. The number of zeros to pad must be **even** — specifically `2k` where `k = scale + work_prec` — because the identity $\sqrt{N \times 10^{2k}} = \sqrt{N} \times 10^k$ only holds when the exponent is even (so that 10^{2k} is itself a perfect square). The final `insert_decimal` shifts the decimal point back by exactly `k` positions.

Irrational / infinite decimals: most square roots (e.g. `sqrt(2)`) produce infinitely many non-repeating decimal digits. Since Newton's iteration runs on a finite-precision integer, it naturally converges to the floor of the true root at the working precision. The result is then hard-capped by `format_fixed_precision` to exactly the user's precision, with trailing digits discarded and the boundary digit rounded half-up.

Precision cap: the internal working precision is capped at 50 decimal places regardless of user-set precision. The final result is then formatted by `format_fixed_precision`. This prevents buffer overflow when the user sets a very large precision.

2.5.2 `abs(x)` — String Prefix Strip

Implemented inline in `evaluate_expression()` (lines ~1425–1430) with a single operation:

```
if (arg[0] == '-') memmove(arg, arg + 1, strlen(arg));
```

No dedicated math function is needed — absolute value on a BigInt string is simply removing the leading `-` character.

2.6 Output Formatting Subsystem (Scientific Notation)

To support scientific notation without polluting the core mathematical algorithms (which function purely on absolute strings), the formatter is implemented as a totally independent post-processing layer: `apply_scientific_format()`.

Zero-Pollution Architecture:

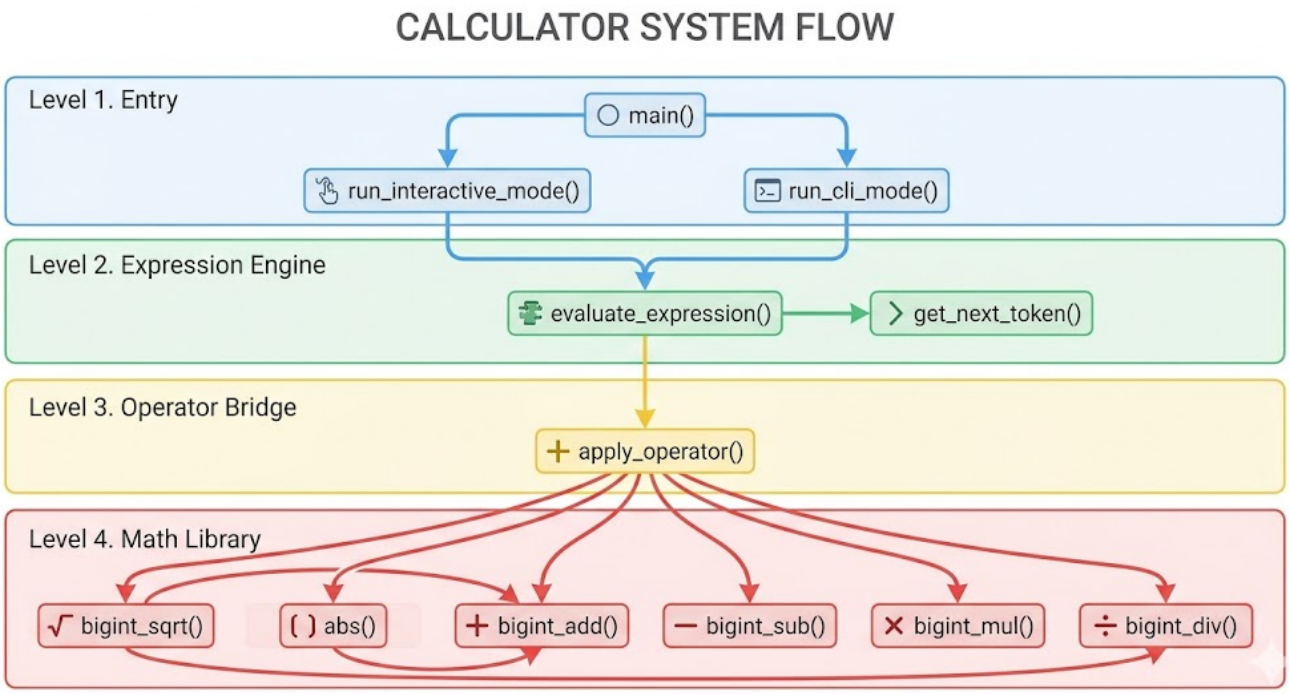
1. All math is strictly calculated in fixed-point format first, up to `precision` (or clamped by `-p max`).
2. Right before printing, `apply_scientific_format` scans the raw string result to calculate its true exponent E.
3. Based on the selected `NotationMode` (`AUTO`, `FIXED`, or `SCI`), it decides whether a transformation is needed.
    - If `AUTO` and E is within the comfortable reading range (e.g. $E \in [-4, 6)$ by default), it skips scientific conversion, perfectly mimicking C's intelligent `%g` specifier.
    - Otherwise, it seamlessly extracts the first non-zero digit, places a decimal point, strips the rest into a mantissa string, rounds the mantissa using `-s` (scientific precision) via `format_fixed_precision`, and appends `e+XX` or `e-XX`.

> [!NOTE] **Dependency on Calculation Precision (`-p`)** Because the formatting layer only triggers *after* the core math engine yields a fixed-point string, the fixed-point string must contain the non-zero digits for the formatter to find them. If you compute `1 / 1000000000000` with the default `-p 6`, the math engine outputs `0`, and the formatter will simply echo `0`. **Therefore, if you want to capture infinitesimal mathematical results like `1.0e-12`, you must set `-p` (Calculation Precision) high enough (e.g., `-p 15`) to capture the non-zero digits before the scientific formatter can convert them.**

This decoupling allows `-x` and `-s` to manipulate visual presentation identically across both interactive and CLI modes without altering the underlying strict BigInt truth.

3. Code Structure

The codebase is organized into four distinct layers. Each layer only calls downward.



Layer	Functions	Responsibility
Entry	`main`, `run_interactive_mode`, `run_cli_mode`	Parse CLI args, read user input, dispatch
Expression Engine	`evaluate_expression`, `get_next_token`	Lexer + Shunting-Yard algorithm
Operator Bridge	`apply_operator`	Strips decimals, aligns scales, dispatches to BigInt, restores result
Math Library	`bigint_add/sub/mul/div`, `bigint_sqrt`, `abs`	Pure arithmetic on arbitrary-precision strings

4. Checking Illegal Input

The code handles illegal input through a rigorously designed multi-layered architecture. Errors are intercepted at three main stages: Lexical Analysis, Syntax Parsing, and Mathematical Execution. All possible errors are defined in the `ErrorCode` enum, allowing the program to safely bubble them up without crashing.

4.1 Error Code Definition

At the beginning of the codebase, the `ErrorCode` enum uniformly manages error types:

- `SUCCESS`: Successful calculation
- `ERR_DIV_BY_ZERO`: Division by zero
- `ERR_INVALID_INPUT`: Invalid input format (e.g., syntax or lexical errors)
- `ERR_INPUT_TOO_LONG`: Input exceeds maximum buffer capacity
- `ERR_UNSUPPORTED_OP`: Unsupported operator
- `ERR_NEGATIVE_SQRT`: Square root of a negative number

4.2 Error Interception Layers

A. Lexical Analysis (Rejecting invalid characters)

This phase is handled by `get_next_token()` and its underlying helper `parse_number_token()`. It validates the characters themselves:

- **Malformed Numbers**: If a number string contains invalid formats (e.g., multiple decimal points, invalid scientific notation, or alphabet letters mixed in), `parse_number_token` throws `ERR_INVALID_INPUT`.
- **Standalone / Invalid Signs**: Explicit positive signs in a unary context (e.g., `+5`) are rejected. Unrecognized mathematical symbols (like `@` or `#`) cause `get_next_token` to return `TOKEN_ERROR`.
- **Unknown Functions**: If a sequence of letters does not match the hardcoded `sqrt` or `abs` functions, the code will eventually mark it as an invalid input.

B. Syntax Parsing (Rejecting invalid expression structures)

This phase is handled by `evaluate_expression()` using the **Shunting-yard algorithm** to check the structural validity of the evaluated string:

- **Token-level Interception**: If `get_next_token()` yields an error token, the parser immediately aborts and returns `ERR_INVALID_INPUT`.
- **Mismatched Parentheses (Extra Right)**: When an extra right parenthesis `)` forces the code to pop operators from a stack, and the stack becomes empty before finding a left parenthesis `(`, it returns `ERR_INVALID_INPUT`.
- **Mismatched Parentheses (Extra Left)**: When the expression ends and the operator stack is flushed, any remaining `(` indicates unmatched parenthesis, triggering `ERR_INVALID_INPUT`.
- **Missing Operands**: If the program pops numbers from the value stack to apply an operator (e.g. evaluating `* 5` or `5 +`) and finds the stack empty, the underlying stack functions return `ERR_INVALID_INPUT`.
- **Standalone / Leftover Numbers**: After the entire expression is processed, the value stack must contain exactly **one** final result. If multiple numbers remain (e.g., a user input `5 5` without an operator), the code traps it and returns `ERR_INVALID_INPUT`.

C. Mathematical Constraints (Rejecting invalid math logic)

Mathematically forbidden operations are intercepted at the underlying BigInt math functions:

- **Division by Zero**: `bigint_div` checks if the divisor is composed entirely of zeros using `is_all_zeros_number()` and throws `ERR_DIV_BY_ZERO`.
- **Negative Square Root**: `bigint_sqrt` checks if the input starts with a negative sign (`-`) and throws `ERR_NEGATIVE_SQRT`.

4.3 Unified Error Output Presentation

Regardless of which layer threw the error, the `evaluate_expression()` function bubbles the `ErrorCode` up. It is eventually caught by the `print_result()` function, which acts as a centralized error dispatcher. For instance, receiving `ERR_INVALID_INPUT` prompts the user with: `The input cannot be`

`interpret as numbers!` This ensures that the user is always presented with human-readable feedback and the program cycle continues smoothly without abrupt termination.

4.4 Memory Safety and Buffer Protection

When dealing with arbitrary-precision arithmetic, uncontrolled fractions (like `1/3`) can easily result in buffer overflows (`Segmentation Fault`) if a user requests unreasonable precision. The system fundamentally protects memory at two levels:

1. **Dynamic Generation Checks**: Inside the core expansion loops (e.g., the decimal expansion loop of `bigint_div`), the remaining capacity `MAX_BIGINT_LEN` is strictly monitored. If the result strings approach the array capacity boundary, the loop safely cuts off and throws `ERR_INPUT_TOO_LONG`.
2. **Safe Precision Ceilings**: The `-p max` shortcut automatically binds the precision to `LIMIT_PRECISION (MAX_BIGINT_LEN - 1024)`, guaranteeing that the system leverages the maximum possible processing space while leaving exactly enough trailing buffer to safely perform carrying, formatting, and mathematical maneuvering without crashing.